Inter-Annotator Reliability Analysis — Power Distance Bias Stream

An Intra-Class Correlation Coefficient (ICC(2,1)) was calculated to assess inter-rater reliability for the Power Distance bias stream using a two-way random, absolute-agreement model. The resulting ICC value of 0.959, with a 95% confidence interval between 0.946 and 0.971, indicates excellent agreement among raters. In line with conventional reliability interpretations (Koo & Li, 2016), this result suggests that annotators demonstrated highly consistent application of the rubric when evaluating model outputs for hierarchical or egalitarian framing.

The pairwise Pearson correlation matrix shows strong correlations across most annotator pairs, generally in the 0.7–0.9 range. These results demonstrate substantial alignment in the absolute values assigned to each item. Minor gaps and slightly lower correlations for a small subset of raters likely reflect uneven task coverage or nuanced differences in how individual annotators perceived expressions of authority or deference within the text.

Rank-based analyses further support these findings. Both Spearman's ρ and Kendall's τ show consistently high correlations ($\rho \approx 0.7$–0.9, $\tau \approx 0.6$–0.8) across overlapping rater pairs. This indicates strong ordinal agreement, meaning raters generally ranked model responses similarly along the low–high power distance continuum.

Taken together, the results indicate that the Power Distance bias stream achieved excellent inter-annotator reliability. Annotators not only demonstrated consistent absolute scoring but also coherent relative judgments of hierarchy and deference cues across model outputs. This high level of reliability underscores the clarity and robustness of the bias rubric for assessing sociocultural hierarchy orientations in model responses.